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Forecasting stock market with nanophotonic reservoir computing system based on silicon optomechanical oscillators

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Abstract: The essence of stock market forecasting is to reveal the intrinsic operation rules of stock market, however it is a terribly arduous challenge for investors. The application of nanophotonic technology in the intelligence field provides a new approach for stock market forecasting with its unique advantages. In this work, a novel nanophotonic reservoir computing (RC) system based on silicon optomechanical oscillators (OMO) with photonic crystal (PhC) cavities for stock market forecasting is implemented. The long-term closing prices of four representative stock indexes are accurately forecast with small prediction errors, and the forecasting results with distinct characteristics are exhibited in the mature stock market and emerging stock market separately. Our work offers solutions and suggestions for surmounting the concept drift problem in stock market environment. The comprehensive influence of RC parameters on forecasting performance are displayed via the mapping diagrams, while some intriguing results indicate that the mature stock markets are more sensitive to the variation of RC parameters than the emerging stock markets. Furthermore, the direction trend forecasting results illustrate that our system has certain direction forecasting ability. Additionally, the stock forecasting problem with short listing time and few data in the stock market is solved through transfer learning (TL) in stock sector. The generalization ability (GA) of our nanophotonic reservoir computing system is also verified via four stocks in the same region and industry. Therefore, our work contributes to a novel RC model for stock market forecasting in the nanophotonic field, and provides a new prototype system for more applications in the intelligent information processing field.

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1. Introduction

Stock market is inextricably linked to the financial assets of the states, enterprises and individuals, while its movement tendency is impacted by various macroeconomic and microeconomic factors as diverse as political climate, interest rates, market news, fiscal policy, commodity price index, investor psychology [1,2]. Accordingly in this sense, the prediction of stock market trend is an extremely arduous challenge due to the stock market is generally noisy, complex, nonlinear, nonparametric, and nonstationary in nature [3,4]. Conventional stock market prediction is commonly based merely on economics, finance and simple mathematical statistical analysis. Nevertheless, with the vigorous advance of artificial intelligence (AI) related technologies and the soaring demand for stock market prediction, the stock market forecasting manipulated by intelligent technologies has been evolved into a research upsurge in economic and financial fields during the latest years.

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The stock market prediction schemes based on intelligent technologies typically models the historical data of stock price as time series, and deems the assumptions that historical tendency of price repeats itself consistently and the stock price data contain all the fundamental factors influencing the movements of stock market, which eludes the evaluation of empirical economic factors [5,6]. According to such comprehension, stock market prediction systems with a variety kind of intelligent technologies has been proposed in succession, such as artificial neural networks (ANN) [7,8], convolutional neural network (CNN) [9,10], recurrent neural network (RNN) [11], support vector machine (SVM) [12,13], stochastic neural network (SNN) [14], and long-short term memory (LSTM) networks [15]. However, there are more or less problems and defects in exploiting a single intelligent technology to predict stock price data with different characteristics, including local optimization, overfitting the learning model and parameter selection difficulties [6,16]. To remedy these drawbacks, a batch of technicians adopt hybrid mechanism and ensemble method to synthesize two or more intelligent technologies for enhancing prediction capabilities and generalization performance [6,17–20]. However, the success of these intelligent technologies in predicting the stock market commonly acquires to configure multilayer connection weights, massive neurons and astonishing amount of hyper-parameters [21,22].

Instead of training vast weights in multiple layers, the reservoir computing (RC) as a particular RNN model only requires to train the output weights exploiting practicable optimization techniques while maintaining the weights of input layer and reservoir internal connection fixed randomly [21,23]. Therefore, the RC can map input signals to high-dimensional space with simpler learning algorithm, lower calculation power and faster learning process to complete prediction and classification tasks in electronics, photonics, spintronics, mechanics, biology and other fields [22–25]. Among them, the photonics-based implementation is one of the most perspective fields for constructing high efficient RC owing to thoroughly utilize the advantages of optical intrinsic properties, containing high data processing speed, large bandwidth, inherent parallelism, and low energy consumption [24,26]. On this account, a mass of ingenious approaches for implementing RC have been gradually proposed based on both spatially distributed and time delayed photonic nodes [24,25,27].

The RC composed of spatially distributed nonlinear photonic nodes allows for a variety of connection topologies using integrated photonic devices on chip, standard diffractive optical elements (DOE), and excitable photonic devices, along with reflecting its parallel processing capability by dealing with large-scale input data without immolating calculation velocity [24,26–32]. Another RC is constituted by a single nonlinear node with one or more time-delayed optoelectronic or all-optical feedback for obtaining time-multiplexed virtual nodes in the temporal domain, in which the nonlinearity alternatively originates from either the discrete element in the feedback loop or the laser response as the feedback delay signal reenters its cavity [24,25]. A number of practical implementations based on vertical cavity surface emitting lasers (VCSEL), semiconductor optical amplifier (SOA), semiconductor ring laser (SRL), semiconductor nanolaser (SNL), and Fabry-Perot laser (FPL) etc. fall into this time-delayed RC [33–37]. Compared to these conventional semiconductor laser (SL), the silicon optomechanical oscillators (OMO) characterized by high quality factor and small mode volume have some unique advantages of small device size, low cost, low power consumption, high scalability, good complementary metal oxide semiconductor (CMOS) fabrication compatibility, favourable monolithic integration and mass production [38–42]. Furthermore, diverse nonlinear dynamics of silicon OMO derived from the strong optomechanical coupling and optical confinement have been successfully observed in recent works [43–46]. In these cases, the above advantages and nonlinearity of the silicon OMO make these oscillators good potential candidates for implementing RC. However, to the best of our knowledge, we have noted that the investigations about nanophotonic reservoir computing based on silicon OMO remained scarce, and especially there are no reports on its application in forecasting stock market.

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In this paper, a novel nanophotonic reservoir computing system based on silicon OMO with photonic crystal (PhC) cavities for stock market forecasting is implemented. Firstly, we introduce the time-delayed feedback mechanism and the injection signal affected by the stock market information into the silicon-based OMO to construct the nanophotonic reservoir computing system with competence in predicting stock market. In order to ensure the accuracy of prediction, the original stock market data is normalized, and then the normalized data is denoised by discrete wavelet transform (DWT). Four globally representative stock indices are selected, and their data are collected for the training and testing process of the forecasting task. This paper presents the long-term prediction results of these four stock indices, and reflects the prediction performance through four relevant error evaluation metrics. On the basis of above, we study the impact of concept drift on the prediction performance of four stock indices over time, and depict the corresponding conclusions. Moreover, the influence of injected light wavelength and injected light power of silicon OMO on the prediction performance is demonstrated by the mapping diagrams, while the effect of time-delayed feedback intensity and injection signal intensity of silicon OMO on the prediction performance is also studied via the mapping diagrams. According to the latest data updated every day, we continuously track and forecast the four stock indices in the one recent year, and compare the daily forecast results with the real values of the stock market, together with statistically analyzing the consistency of their rising and falling trends. Moreover, we explore the capacity of transfer learning (TL) in nanophotonic reservoir computing system based on silicon OMO, i.e., embodied by using the massive data accumulated for a long time in the stock sector to predict the stock prices of several companies with short listing time under this sector. This has important strategic significance and practical value for predicting stocks with short issuance time and less time series data. Ultimately, we discuss the generalization ability (GA) of nanophotonic reservoir computing system based on silicon OMO, which is reflected by training any one of the four selected stocks in the same region and industry to predict the other three stocks.

The remainder of this paper is organized as follows. In Section 2, the system scheme is introduced in detail. In Section 3, the selection and setting of stock market data associating with the measurement indicators of its prediction performance are illustrated. In Section 4, the forecasting results and the influence of system parameters are presented. In Section 5, the transfer learning and generalization ability of stock prediction system are explored and discussed. Section 6 summarizes the results and provides advices for further research.

2. System scheme

The schematic diagram of nanophotonic reservoir computing system based on silicon OMO with PhC cavities for forecasting stock market is elaborated in Fig. 1. The whole system is comprised of three main parts: the input layer, reservoir, and output layer. In the input layer, the original stock market data is collected and recorded as $u(n)$ with a period of time duration T, where n is the discrete time index. Henceforth the $u(n)$ is transformed into preprocessed stock market data $U(n)$ through pretreatment operations of normalization and wavelet denoising. A chaotic mask signal $M(t)$ with identical time duration T generated by our previous work [47] is exploited to guarantee the variability of input signal over the distinct virtual nodes where the information is read out [33,48]. On this foundation, the $U(n)$ is multiplied by $M(t)$ and rescaled within appropriate range to acquire the masked input signal $S(t)$, thereafter the $S(t)$ is transmitted into the reservoir via modulation. In essence, the stock price information of input signal is hidden in the amplitude of masked signal per period after the input layer processed. In the reservoir, the injection laser (I-Laser) is utilized as the input light source to stimulate the silicon OMO. And the physical fabrication parameters corresponding to the silicon OMO have been elaborated in literatures [43] and [39]. The reservoir is a master-slave configuration constructed by silicon OMO injected by a drive laser (D-Laser), where the output signal of I-Laser and the modulated signal of D-Laser

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enter the silicon OMO as one signal through the beam combiner (BC). The states with a time interval θ output by silicon OMO constitutes the virtual network nodes. There are L virtual nodes at each interval θ in the feedback loop with a delay time τ. This feedback delay time τ nearly matches the input time duration T ($= L\theta$) in the case of de-synchronization scheme, i.e., $\tau = T + \theta$ [23,33,34]. In the output layer, all the states of virtual nodes $x(n)$ corresponding to the transient response of silicon OMO are read out for training the output weight W_{out}, and the ultimate output $y(n)$ are weighted and linearly summed up, i.e., $y(n) = \sum x(n)W_{out}$. Notably, the optimal output weights are calculated by exploiting linear least-squares method in this paper for minimizing the mean-square error between the anticipant target values and RC output values during the training stage.

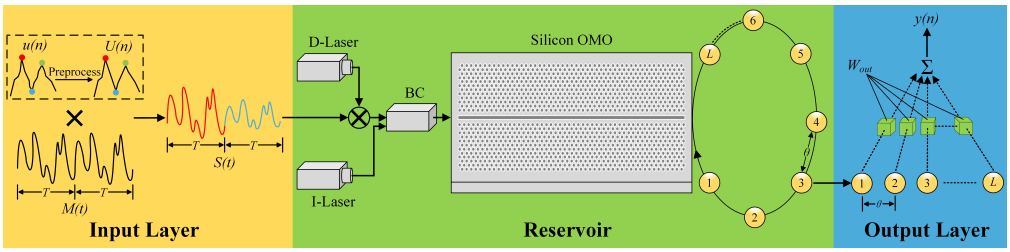


Fig. 1. The schematic diagram of nanophotonic reservoir computing system for forecasting stock market based on silicon OMO with time-delay feedback loop. The processing scheme consists of the input layer, reservoir, and output layer. $u(n)$: original stock market data; $U(n)$: preprocessed stock market data; $M(t)$: chaotic mask signal; T: input holding time; $S(t)$: masked input signal; D-Laser: drive laser; I-Laser: injection laser; BC: beam combiner; θ: virtual node interval; W_{out}: trained output weight; $y(n)$: RC output.

With regard to the nanophotonic reservoir computing system implemented in this work, we model the nonlinear dynamics of the silicon OMO under feedback loop and information injection by considering the OMO and several strong nonlinear effects as diverse as the two-photon absorption (TPA), free-carriers absorption (FCA) and thermo-optic effect [40,41,43,44]. Consequently, the entire operation process in the reservoir can be modeled as follows:

$$\begin{aligned}\frac{dE}{dt} =& -\frac{1}{2}(\gamma_i + \gamma_e + \frac{\Gamma_{TPA}\beta_{si}c^2}{V_{TPA}n_g^2}|E(t)|^2 + \frac{\sigma_{si}cN(t)}{n_g})E(t)\\ &+ i(-g_0\sqrt{\frac{2m_{eff}\Omega_m}{\hbar}}x(t) + \frac{\omega_0}{n_{si}}(\frac{dn_{si}}{dT}\Delta T(t) + \frac{dn_{si}}{dN}N(t)) + \delta_\omega)E(t)\\ &+ \sqrt{\gamma_e(P_{in} + k_{fb}E(t-\tau) + k_{inj}E_{inj}(t))}\end{aligned} \tag{1}$$

$$\frac{d^2x}{dt^2} + \Gamma_m\frac{dx}{dt} + \Omega_m^2 x(t) = \frac{g_0}{\omega_0}\sqrt{\frac{2\Omega_m}{\hbar m_{eff}}}|E(t)|^2 \tag{2}$$

$$\frac{dN}{dt} = -\frac{N(t)}{\tau_{fc}} + \frac{\Gamma_{FCA}\beta_{si}c^2}{2\hbar\omega_0 n_g^2 V_{FCA}^2}|E(t)|^4 \tag{3}$$

$$\frac{d\Delta T}{dt} = -\frac{\Delta T(t)}{\tau_{th}} + \frac{\Gamma_{PhC}}{\rho_{si}c_pV_{PhC}}(\gamma_i + \frac{\Gamma_{TPA}\beta_{si}c^2}{V_{TPA}n_g^2}|E(t)|^2 + \frac{\sigma_{si}cN(t)}{n_g})|E(t)|^2 \tag{4}$$

where E is the intracavity electric-field amplitude, x is the motional displacement, N denotes the free-carrier density, and ΔT represents the cavity temperature variation. δ_ω is the angular frequency detuning, i.e., $\delta_\omega = \omega_L - \omega_0$, where ω_L and ω_0 are the angular frequency of injection

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light and photonic crystal cavity resonance respectively, and $\xi_w = \delta_\omega/\gamma_i$ is the normalized injection detuning. P_{in} is the injected optical power, k_{fb} and τ are the feedback strength and time delay of feedback loop. k_{inj} denotes the injection strength from the D-Laser to Silicon OMO, and the $E_{inj}(t)$ denotes the slowly varying complex electric field of the D-Laser. In consideration of modulating the masked input signal $S(t)$ with D-Laser, the $E_{inj}(t)$ can be depicted as following equation [23,33]:

$$E_{inj}(t) = \sqrt{I_d} e^{(i\pi S(t))} \tag{5}$$

where the I_d is the output intensity of D-Laser, and the masked input signal $S(t)$ can be expressed as:

$$S(t) = M(t) \times U(n)\gamma \tag{6}$$

where $M(t)$ is the chaotic mask signal with periodicity T, $U(n)$ is the preprocessed stock market data with time duration T, and the γ is the scaling factor.

Table 1. Modeling parameters.

Symbol	Physical meaning	Value
γ_i	Linear loss	19 GHz
$\gamma_e/2\pi$	External coupling rate	2.2 GHz
Γ_{TPA}	TPA confinement factor	0.8012
β_{si}	TPA coefficient	8.4×10^{-12} m/W
V_{TPA}	TPA mode volume	6.4×10^{-19} m^3
n_g	Group index	3.476
σ_{si}	FCA cross section	1×10^{-21} m^2
g_0	Vacuum optomechanics coupling strength	690 kHz
m_{eff}	Mechanical osc. effective mass	2.4×10^{-14} kg
$\Omega_m/2\pi$	Mechanical frequency	110 MHz
λ_0	Resonance wavelength	1572.8 nm
λ_L	I-Laser resonance wavelength	1572.8 nm
n_{si}	Refractive index of Si	3.476
dn_{si}/dT	Thermo-optic coefficient	1.86×10^{-4} K^{-1}
dn_{si}/dN	Free carrier coefficient	-1.73×10^{-27} m^3
P_{in}	Injected optical power	1.3×10^{-3} W
L	Virtual node number	100
T	Input time duration	1 ns
I_d	D-Laser intensity	6.56×10^{20}
γ	Scaling factor	1
$\Gamma_m/2\pi$	Mechanical dissipation rate	110 kHz
τ_{fc}	Free-carrier lifetime	150 ps
Γ_{FCA}	FCA confinement factor	0.79
V_{FCA}	FCA mode volume	6.9×10^{-19} m^3
τ_{th}	Thermal dissipation life time	9.7 ns
Γ_{PhC}	Thermal confinement factor	0.769
ρ_{si}	Density of material Si	2.33×10^3 kg/m^3
c_p	Specific heat capacity	700 J/(kg·K)
V_{PhC}	Thermal mode volume	1×10^{-18} m^3

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The aforementioned equations (1)–(4) can be solved by utilizing fourth-order Runge-Kutta algorithm in numerical simulation, while the physical meaning and value of reservoir parameters are shown in Table 1 unless otherwise specified.

Moreover, the original stock market data $u(n)$ can be normalized in a desired range as follows [1]:

$$u_i' = a \times \frac{u_i - u_{min}}{u_{max} - u_{min}} \pm b \tag{7}$$

where u_i' is the normalized stock market data ($i = 1, 2, \ldots, n$), u_{min} and u_{max} are the minimum and maximum value of the original data, respectively. Here, a and b are two coefficients to make the preprocessed data in a desired range, here we set $a = 1$ and $b = 0$ in the work.

Meanwhile, the DWT algorithm is adopted for denoising the normalized data in the pretreatment process, since it can flexibly handle the non-stationarity and highly irregular stock market data by time and frequency scale decomposition [3]. This work utilizes the computationally efficient Daubechies (db3) wavelet with three level decomposition as the DWT tool. In addition, it merits our attention that the first one hundred stock market data are left out in the RC training process as the initialization data for purpose of stabilizing its performance and also washing out the initial memory of the RC system.

3. Stock market data and evaluation metrics

3.1. Stock market data selection and setting

In the stock market, the overall level of various stock prices and their variations are generally reflected by the stock indices, which can provide the dynamic information and development form of the stock market. The maturity of stock market is determined by multiple factors such as operation mechanism, market scale, competition degree and order degree, thus different market maturities may affect the accuracy of the prediction model. After several hundred years of successive development, European and American stock markets have become mature stock markets with sophisticated market operation system, huge market scale and reasonable supervision institution. In contrast, the stock market in mainland China has a relatively short development time while the market operation and supervision system are imperfect. Although it is developing rapidly, the stock market in mainland China is still in the developing stage and is an emerging stock market. In this work, we select two of the most representative stock indices in mainland China to delegate the emerging stock market, namely Shanghai Stock Exchange Composite Index (SHSECI, stock symbol (SS): 000001) and Shenzhen Stock Exchange Component Index (SZSECI, SS: 399001). Besides, we also select the National Association of Securities Dealers Automated Quotations (NASDAQ) Composite Index (NASDAQCI, SS: IXIC), one of the most representative indexes in Americas, and the London Financial Times Stock Exchange 100 Index (LFTSE100I, SS: FTSE), one of the most representative indexes in Europe, as the delegate stock indexes of mature stock market. The historical daily closing prices of these four representative stock indexes are collected from the public financial data services of Royalflush client software provided by Hithink Royalflush Information Network Co.,Ltd. Details of the four selected indexes are presented in Table 2.

Table 2. Stock market data settings.

Stock Index	Dataset Period	Trading Days	Training-dataset Period	Training Days	Test-dataset Period	Test Days
SHSECI	06/01/1992-25/12/2020	7073	19/07/2004-18/11/2016	3000	21/11/2016-25/12/2020	1000
SZSECI	06/01/1992-25/12/2020	7062	19/07/2004-18/11/2016	3000	21/11/2016-25/12/2020	1000
NASDAQCI	06/01/1992-24/12/2020	7293	27/01/2005-03/01/2017	3000	04/01/2017-24/12/2020	1000
LFTSE100I	06/01/1992-24/12/2020	7332	01/03/2005-12/01/2017	3000	13/01/2017-24/12/2020	1000

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As illustrated in Table 2, all four stock indexes roughly span twenty nine years from 6th January 1992 to 24th/25th December 2020, with 7073, 7062, 7293, and 7332 trading days respectively. The total volume of 28760 stock index data over a long period of time is large enough to accommodate almost all sorts of market conditions. The last 1000 data from 21st November 2016 (4th January 2017, 13th January 2017) to 25th December 2020 (24th December 2020) are used as the test dataset. And the previous 3000 data before the test dataset from 19th July 2004 (27th January 2005, 1st March 2005) to 18th November 2016 (3rd January 2017, 12th January 2017) are used as the training dataset. The amount of data in the training set and test set will not change unless otherwise specified. It's worth noting that the remaining data of the overall dataset except the training set and test set will be occupied in the subsequent section of concept drift.

3.2. Evaluation metrics of prediction performance

Prediction performance can usually be measured by evaluating the error between the predicted value and the real value. In this work, the normalized mean square error (NMSE), root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) are employed to evaluate the prediction error, which can be calculated by the following equations [1,3,4,21,33]:

$$NMSE = \frac{1}{H}\frac{\sum_{n=1}^{H}(y(n)-\bar{y}(n))^2}{var(y(n))} \tag{8}$$

$$RMSE = \sqrt{\frac{1}{H}\sum_{n=1}^{H}(y(n)-\bar{y}(n))^2} \tag{9}$$

$$MAE = \frac{1}{H}\sum_{n=1}^{H}|y(n)-\bar{y}(n)| \tag{10}$$

$$MAPE = \frac{1}{H}\sum_{n=1}^{H}\left|\frac{y(n)-\bar{y}(n)}{y(n)}\right| \tag{11}$$

where $y(n)$ and $\bar{y}(n)$ are the real value and the forecast value, respectively. H represents the total number of samples, and the *var* indicates the variance. In general, the predictive performance of a RC system can be considered good when the NMSE is below 0.1 [33].

Furthermore, the consistency of stock fluctuation trend direction between forecast value and real value can be evaluated by the directional symmetry (DS), which is expressed as the following equation [4]:

$$DS = \frac{1}{H}\sum_{n=1}^{H}\mu_n, \quad \mu_n = \begin{cases} 1, & [y(n+1)-y(n)][\bar{y}(n+1)-\bar{y}(n)]>0 \\ 0, & others \end{cases} \tag{12}$$

4. Forecasting results and discussion

4.1. Forecasting results and comparative analysis

In accordance with the system scheme illustrated in Fig. 1, the four stock indexes are trained in training-dataset period of Table 2, afterwards the four stock indexes are tested in test-dataset period. In order to make the prediction performance of each stock index better, we configure different parameter values for four stock indexes respectively, owing to the stock markets represented by the four stock indexes have different market characteristics and the identical parameter values cannot meet the four stock indexes at the same time. Following the modeling parameters in Table 1, the different parameter values configured for each four stock indexes are shown in Table 3.

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Table 3. Parameter values for each of the four stock indexes.

Stock Index	k_{fb}	k_{inj}
SHSECI	1×10^{4}	1×10^{-13}
SZSECI	1×10^{4}	2.8×10^{-13}
NASDAQCI	9.6×10^{3}	1×10^{-15}
LFTSE100I	1×10^{4}	9.3×10^{-15}

Figures 2(a)-(d) successively demonstrate the forecasting results of SHSECI, SZSECI, NASDAQCI, and LFTSE100I in test-dataset period, where the red lines represent the real values and the blue lines denote the forecast values respectively. As can be seen from all the four subgraphs in Fig. 2, the forecast blue curves are highly consistent with the real red curves, indicating the tiny errors between the forecast values and real values. Besides, such phenomenon can be found through careful observation, i.e., the forecast blue curves at the point where the real red curves with violent fluctuations usually cannot fit the real red curves very well, thus there are larger errors between the forecast values and real values than the small fluctuation spots. This phenomenon coincides with the actual situation that the more severe of the fluctuation resulting in the worse prediction performance under the real stock market circumstance. Such reasonable and conservative prediction verifies the impact of the anchoring effect on the forecasting system, due to the past historical data is the crucial determinant to the prediction results.

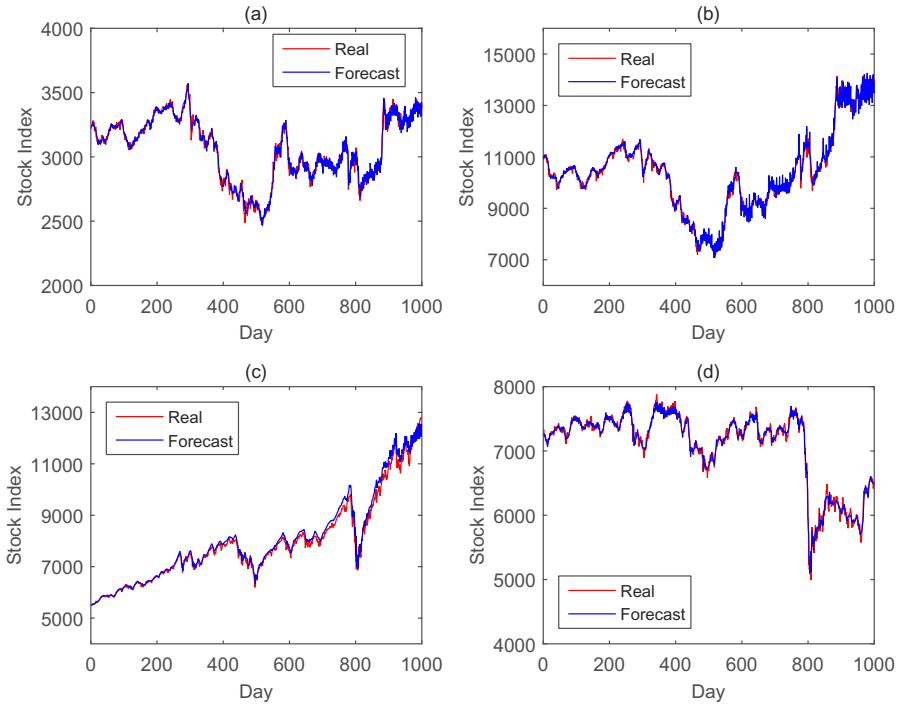


Fig. 2. The forecasting results of four stock indexes in test-dataset period, where the red and blue lines represent the real values and the forecast values respectively. (a) SHSECI; (b) SZSECI; (c) NASDAQCI; (d) LFTSE100I.

For specific quantization errors of four stock indexes in Fig. 2, four relevant error evaluation metrics containing NMSE, RMSE, MAE, and MAPE are calculated in Table 4. The results of our

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work in Table 4 corroborate the fact in Fig. 2 that the forecast blue curves are very close to real red curves, since all the NMSEs, RMSEs, MAEs, and MAPEs are less than 0.024, 0.019, 0.013, and 1.764×10^{-5}, respectively. In the case of similar prediction errors shown in Table 4, the injection strengths k_{inj} of mature stock market represented by NASDAQCI and LFTSE100I are about two orders of magnitude smaller than the emerging stock market represented by SHSECI and SZSECI. Furthermore, the NMSEs of mature stock market are marginally smaller than the emerging stock market, while the opposite results are displayed in RMSEs, MAEs, and MAPEs. The comparison between the parameter configuration of Table 3 and prediction errors of Table 4 reveals that the stock indexes with different market maturity have distinct properties, yet the stock indexes with same market maturity have certain intrinsic relations.

Table 4. Evaluation metrics of forecasting results and comparison results.

Work	Index	NMSE	RMSE	MAE	MAPE
Our Work	SHSECI	0.0232	0.0074	0.0056	5.6857×10^{-7}
	SZSECI	0.0203	0.0132	0.0092	3.8394×10^{-7}
	NASDAQCI	0.0167	0.0183	0.0125	1.7634×10^{-5}
	LFTSE100I	0.0145	0.0161	0.0115	1.6127×10^{-6}
Ref. [1]	SHSECI	N/A	847.5841	675.9543	0.1948
Ref. [3]	LFTSE100I	N/A	31.0000	25.0000	0.0056
Ref. [4]	SHSECI	0.8251	0.0141	0.0092	N/A
	SZSECI	0.9371	0.0172	0.0110	N/A
Ref. [14]	LFTSE100I	N/A	54.2607	43.8524	0.0096
Ref. [21]	SHSECI	N/A	30.7600	22.0700	0.0074
	NASDAQCI	N/A	78.0900	53.2400	0.0076
	LFTSE100I	N/A	54.0200	41.0300	0.0056

To comparatively analyze the prediction system performance implemented in our work, some literatures containing the same research objects (SHSECI, SZSECI, NASDAQCI, and LFTSE100I) and evaluation indicators (NMSE, RMSE, MAE, and MAPE) are added to Table 4. It can be find from the comparison results in Table 4, the prediction errors of Ref. [3] is close to the Ref. [14] and Ref. [21]. The work of Ref. [3], Ref. [14] and Ref. [21] have all achieved good stock index prediction performance, and our work is fortunate to have smaller prediction errors while obtaining good stock index prediction results. Additionally, the RMSEs and MAEs of Ref. [4] are very close to the prediction errors of our work, however its NMSEs are much higher than the NMSEs of our work. As a consequence, these contrastive analyses show that our work also has a powerful prediction capacity for stock index, and even has smaller prediction errors than the literatures listed in Table 4.

4.2. Concept drift and stock market forecast

Concept drift phenomenon constantly appears in stock market technical analysis owing to the financial time series is a special dynamic data stream, whose potential statistical characteristics change in an unpredictable way over time, resulting in the continuous depreciation of the knowledge learned from historical data analysis, thus reducing the prediction accuracy of the constructed model. Evidently, it is a vital issue to detect and surmount the concept drift in stock analysis in time. However, there are rare studies on maintaining the prediction accuracy of learned model in the operation process.

In this work, the impact of concept drift on forecasting performance is studied in Fig. 3 by changing the training-dataset period and training days in Table 2, together with taking the forecast

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error of 1000 test days in the fixed test-dataset period as the research object. The prediction errors of SHSECI, SZSECI, NASDAQCI, and LFTSE100I are represented by the red, blue, gray, and green curves respectively. The NMSE results of four stock indexes are shown in Fig. 3(a), where the NMSE larger than 1 is set equal to 1, since the NMSE less than 0.1 is regarded as a good prediction result, and an excessively large NMSE is meaningless. For SHSECI depicted by the red curve, the NMSE are all smaller than 0.05 with good forecasting performance when the training day of training-dataset is less than 4000. Nevertheless, the NMSE increases gradually when the training day is greater than 4000, especially in the range of 4500 to 4720, it even shows a step-like increase. As the training day of training-dataset bigger than 4720, the NMSE maintains basically at 1 excepting for the reduction to 0.92 at 4900, leading to a sharp decline in the forecasting performance. For SZSECI depicted by the blue curve, the NMSE are nearly all smaller than 0.1 with a relatively good forecasting performance when the training day is less than 1860. In the range of 1860 to 1960, the NMSE curve rises slightly and drops rapidly to 0.1, afterwards the system temporarily recovers its good forecasting performance until the training day increases to 2340. As the training day continues to increase from 2340, the NMSE curve increases sharply and then decreases swiftly with a peak. After that, the good forecasting performance briefly appears again with NMSE less than 0.1 until training day increases to 3060. From then on, NMSE increased violently, although there are some significant declines after 4740, the NMSE is still all over 0.7, implying very poor prediction effect. For NASDAQCI depicted by the gray curve, the NMSE curve fluctuates frequently and substantially within all the training days, thus it is difficult to capture the relatively stable change within a certain range and meet the good forecast condition of NMSE less than 0.1. For LFTSE100I depicted by the green curve, the NMSE varies dramatically despite there exist a tremendous amount of good forecasting performance cases as training day less than 1880 or more than 3800. Moreover, the NMSE almost always remains below 0.04 when training day varies from 1880 to 3800, hence the system has good predictive performance in this range. However, the relative variation trend of forecast error can not be fully reflected only by the NMSE.

The RMSE results of four stock indexes are shown in Fig. 3(b), the relative errors of the four indexes are effectively controlled within a smaller error range benefiting from root mean square operation. SHSECI has minimum RMSE in the range of training days less than 4000, and the overall growth range of SHSECI is also smaller than SZSECI and NASDAQCI. The range of minimum forecast error for NASDAQCI is also easier to observe from the RMSE compared with NMSE in Fig. 3(a), i.e., the range of training day is from 2340 to 3080. In fact, the forecast error of NASDAQCI is much larger than the other three stock indexes when the training day is greater than 4000. The MAE results of four stock indexes are shown in Fig. 3(c), where the similar variation trends of the forecast errors are separated more clearly. The MAPE results of four stock indexes are shown in Fig. 3(d), in which the closest forecast errors appear in the range of 2440 to 3240. As a result, both too much or less training days make the prediction error large, and this study find that for the 1000 test days in the fixed test-dataset period, smaller forecast errors with good prediction performance are more likely to occur in the range of 2000 to 4000 training days. Analyzed at a deeper level, the results in Fig. 3 expound that the concept drift has a great impact on the prediction performance of stock market. It is difficult to learn the internal change law of current stock market in a short training-dataset period, while the internal change law of stock market learned in a long-term training-dataset period is outdated. By selecting the appropriate training days of training-dataset in the same way as this work can provide solution and suggestion for overcoming the concept drift problem in stock market environment to a certain extent.

4.3. RC parameters and forecasting performance

For the sake of comprehensively exploring the influence of RC parameters on forecasting performance, the mapping diagrams of the forecast errors in the parameter space of (ξ_w, P_{in})

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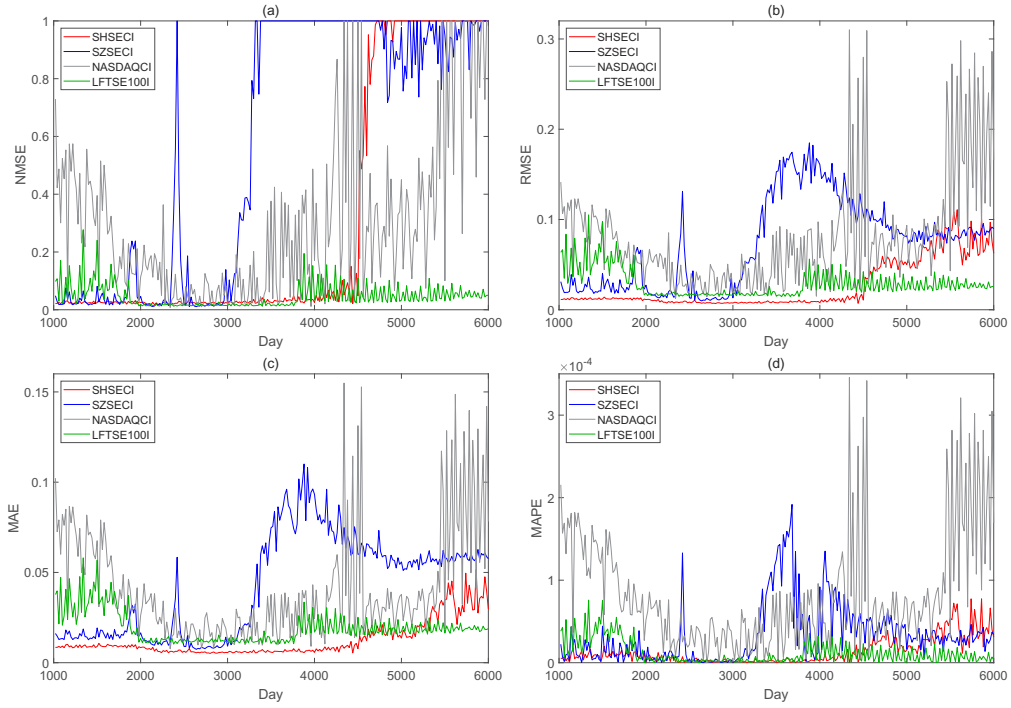


Fig. 3. The impact of concept drift on forecasting performance. The red, blue, gray, and green curves denotes the forecast errors of SHSECI, SZSECI, NASDAQCI, and LFTSE100I, respectively. (a) NMSE; (b) RMSE; (c) MAE; (d) MAPE.

are demonstrated in Fig. 4, where the first to fourth rows respectively represent the results of SHSECI, SZSECI, NASDAQCI, and LFTSE100I, while the first to fourth columns represent the results of NMSE, RMSE, MAE, and MAPE respectively. As can be seen from Figs. 4(a1)-(a4), all the four evaluation metrics of SHSECI are dark blue in the whole area except for the range of ξ_w<1 along with P_{in}>2.6, indicating that the prediction errors are exceedingly small with good forecasting performance in the parameter range. In the region of ξ_w<1 and P_{in}>2.6, the prediction errors gradually increases with the rise of P_{in}, and finally present red high error regions. Through careful observation in Fig. 4(a4), the dark blue becomes slightly lighter in the area around ξ_w = 5 and ξ_w = 15, where the prediction error increases marginally but is still very small. The variation of SZSECI prediction errors presented in Figs. 4(b1)-(b4) are very similar to SHSECI, the difference is that the errors of SZSECI in the region of ξ_w<1 and P_{in}<0.8 with lighter blue are larger than SHSECI, and the errors of SZSECI increase faster with P_{in} rise for bigger red areas than SHSECI. However, the situations of NASDAQCI presented in Figs. 4(c1)-(c4) are utterly distant from SHSECI and SZSECI. In Fig. 4(c1), the red area almost fills the range of P_{in}<1, regardless of the variation with ξ_w. For the region of P_{in}>1, the red regions with high prediction errors mainly appear in the areas where ξ_w are 0, 5, 8, 11, 14 to 20. The variation trends of RMSE, MAE, and MAPE in Figs. 4(c2)-(c4) are consistent with the NMSE in Fig. 4(c1). Therefore, NASDAQCI is sensitive to parameter variation, and it is difficult to capture the dark blue region with high prediction performance in the parameter range. Furthermore, the red area is almost full in the region with P_{in}<0.7 in Fig. 4(d1), while the most significant feature of LFTSE100I is that there is a straight-line area with red high prediction error where P_{in} equals to 1.2 and ξ_w ranges from 4 to 20. Fortunately, instead of a mass of red prediction error regions like NASDAQCI in Fig. 4(c1), vast amounts of dark blue regions

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with lower forecast errors of LFTSE100I appear in the region of $P_{in}>1$. Likewise, the variation tendencies in Figs. 4(d2)-(d4) are in accordance with Fig. 4(d1).

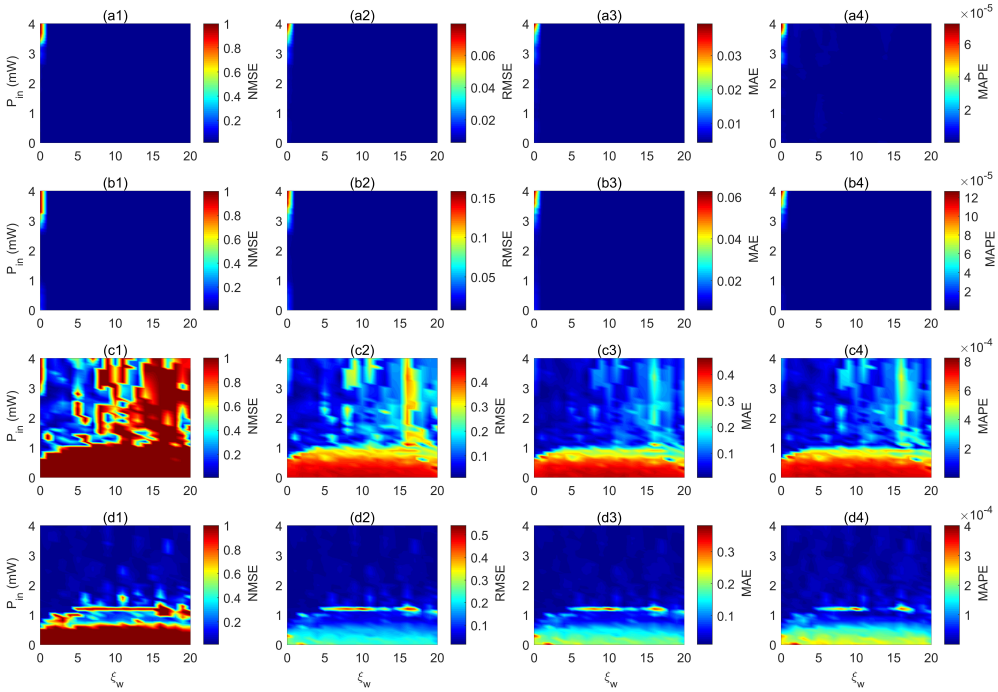


Fig. 4. Mapping diagrams of the forecast errors in the parameter space of (ξ_w, P_{in}), where: (a1)-(a4) SHSECI; (b1)-(b4) SZSECI; (c1)-(c4) NASDAQCI; (d1)-(d4) LFTSE100I.

In addition, the mapping diagrams of the forecast errors in the parameter space of (k_{inj}, k_{fb}) are demonstrated in Fig. 5, where the representative meaning of each letter serial number is the same as Fig. 5. In Fig. 5(a1), the dark blue region with low prediction error mainly appears in the area of $k_{inj}<2.55\times10^{-13}$ and $k_{fb}<1\times10^{4}$. On the other hand, the red region with high prediction error primarily appears in two areas of $k_{inj}<9.5\times10^{-14}$ and $k_{fb}>1.4\times10^{4}$, together with $k_{inj}>9.9\times10^{-14}$ and $k_{fb}>1.1\times10^{4}$. More particularly, when the k_{inj} is greater than 2.61×10^{-13} regardless of the value of k_{fb}, the prediction error is relatively high resulting in a large red area. Even more interesting is that the gradient variation process from the dark blue region with low forecast error to the red region with high forecast error presents an image in Fig. 5(a1) similar to the coastline in geographical map. The changing trends of Figs. 5(a2)-(a4) are consistent with Fig. 5(a1). In Figs. 5(b1)-(b4), the red region with high forecast error mostly appears in the area of $k_{inj}<9.3\times10^{-14}$ and $k_{fb}>1.4\times10^{4}$, together with the area of $k_{inj}>1.09\times10^{-13}$ and $k_{fb}>0.9\times10^{4}$. In most of the remaining parameter space, the forecast errors of the system are extremely small and there are more dark blue areas than Figs. 5(a1)-(a4). In the displayed parameter space of Fig. 5(c1), almost all of its NMSE are in red regions with high forecast errors, while the dark blue regions with low forecast error only exist in the very narrow range of $k_{inj}<9\times10^{-15}$ and $k_{fb}<1.25\times10^{4}$. Besides, more evident relative changes of forecast errors are exhibited in Figs. 5(c2)-(c4), the forecast errors are higher in the range of $6.5\times10^{-14}<k_{inj}<2.31\times10^{-13}$ and $k_{fb}<0.6\times10^{4}$. The changing trend of Figs. 5(c2)-(c4) are in accordance with Fig. 5(c1), and it is also hard to acquire the dark blue region with high prediction performance in the parameter range. In Fig. 5(d1), the red regions are principally distributed in the following four areas. The first distribution red area is the largest, which is in the range

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of $k_{fb}>1.3\times10^{4}$, irrespective of the value k_{inj}. The second distribution red area is in the range of $1.61\times10^{-13}<k_{inj}<2.69\times10^{-13}$ and $0.85\times10^{4}<k_{fb}<1.65\times10^{4}$, and its shape is similar to the head of a trident. The third distribution area is V-shaped sporadically spread in range of $2.5\times10^{-14}<k_{inj}<1.63\times10^{-13}$ and $0.05\times10^{4}<k_{fb}<1.4\times10^{4}$. At last, the fourth distribution red area is in the range of $k_{inj}>1.81\times10^{-13}$ and $k_{fb}<0.6\times10^{4}$. Intriguingly, the gradient variation process from dark blue area to red area shows an image in Fig. 5(d1), which is similar to the archipelago in the geographic map. The changing tendencies of Figs. 5(d2)-(d4) are coincided with Fig. 5(d1).

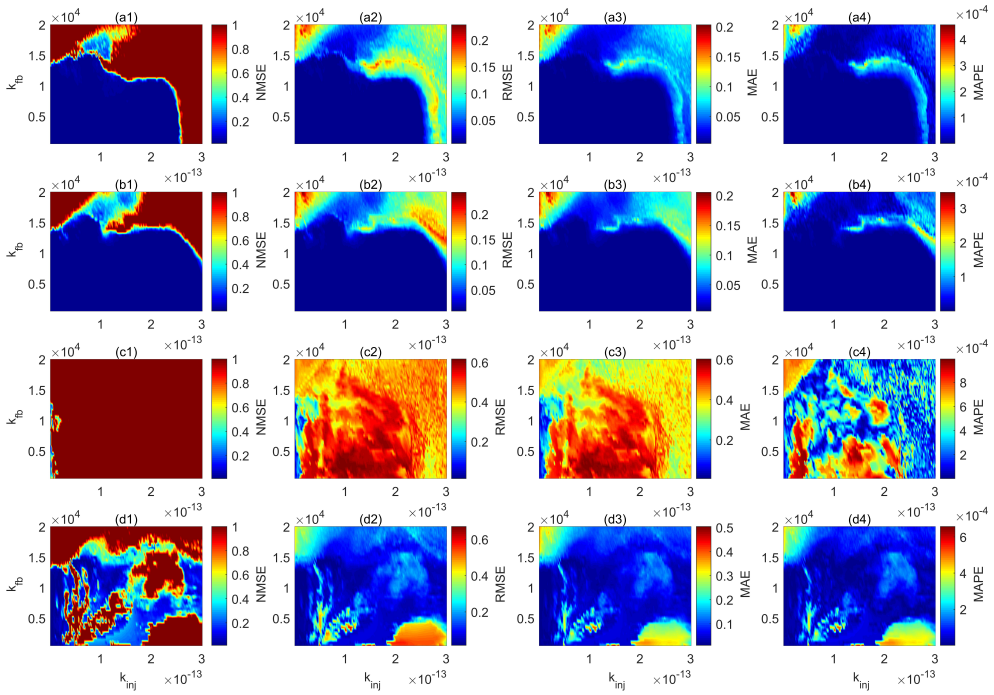


Fig. 5. Mapping diagrams of the forecast errors in the parameter space of (k_{inj}, k_{fb}), where: (a1)-(a4) SHSECI; (b1)-(b4) SZSECI; (c1)-(c4) NASDAQCI; (d1)-(d4) LFTSE100I.

Consequently, the results in Fig. 4 and Fig. 5 once again sufficiently validate that the variation of RC parameters between the emerging stock markets (represented by SHSECI and SZSECI) and the mature stock markets (represented by NASDAQCI and LFTSE100I) complies different variation rules, while the stock indexes with the same stock maturity follow similar variation laws. Furthermore, we find that the mature stock markets are more sensitive to the change of RC parameters than the emerging stock markets.

4.4. *Direction forecast of stock market fluctuation trend*

Focusing on the latest data updated every day in the stock market, we persistently track and forecast the four stock indexes in the one recent year for contrasting the daily forecast results with the real values of the real time stock market, thereafter the consistency of their rising and falling tendency are calculated by DS. The stock market data of the latest one year are specifically listed in Table 5, where the corresponding training-dataset and the overall dataset of each stock index are also enumerated.

In Table 5, the test-dataset period of four stock indexes are uniformly set to the latest one year from 3rd August 2020 to 6th August 2021. To acquire better DS results, we tried different

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Table 5. Stock market data settings for direction forecast of stock market fluctuation trend.

Stock Index	Dataset Period	Trading Days	Training-dataset Period	Training Days	Test-dataset Period	Test Days
SHSECI	01/04/2008-06/08/2021	3250	01/04/2008-31/07/2020	3002	03/08/2020-06/08/2021	248
SZSECI	19/11/2009-06/08/2021	2850	19/11/2009-31/07/2020	2602	03/08/2020-06/08/2021	248
NASDAQCI	24/03/2010-06/08/2021	2857	24/03/2010-31/07/2020	2601	03/08/2020-06/08/2021	256
LFTSE100I	10/09/2010-06/08/2021	2756	10/09/2010-31/07/2020	2499	03/08/2020-06/08/2021	257

training days for four stock indexes and ultimately settled on 3002 days, 2602 days, 2601 days, and 2499 days. The direction forecast results of stock market fluctuation trend for four stock indexes are demonstrated in Fig. 6, and its DS results are manifested in Table 6.

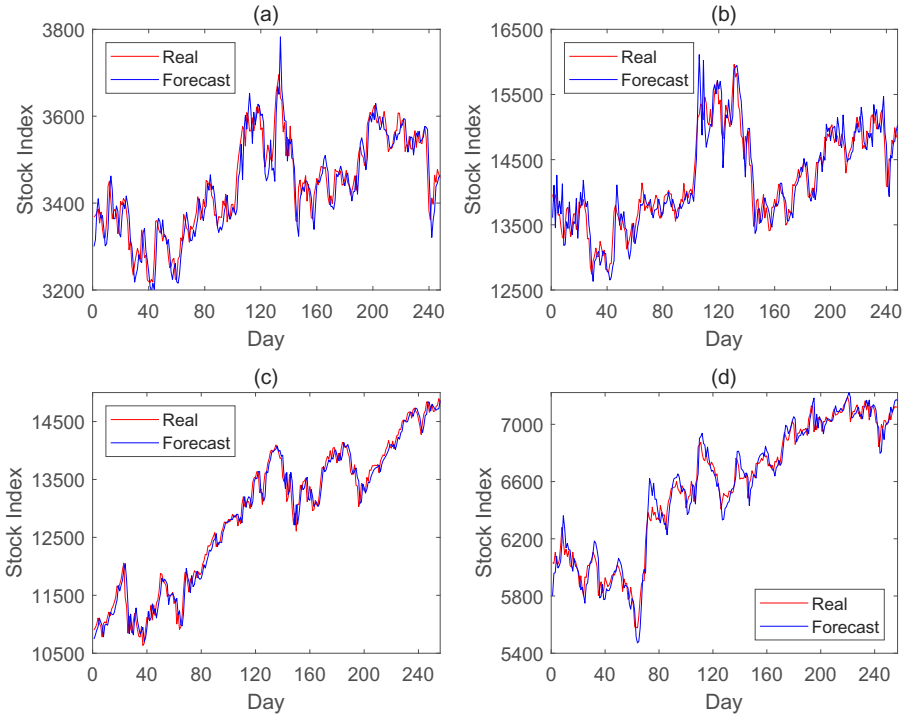


Fig. 6. The forecasting results of four stock indexes for direction forecast of stock market fluctuation trend, where the red and blue lines represent the real values and the forecast values respectively. (a) SHSECI; (b) SZSECI; (c) NASDAQCI; (d) LFTSE100I.

As can be seen from Table 6, the DS results of four stock indexes in our work are 0.5282, 0.5242, 0.5078, and 0.5097, respectively. The DS results of our RC forecasting system are compared with the RNN in Ref. [4] since RC is a special RNN, and it is find that our results of SHSECI and SZSECI are little higher with better direction forecasting performance. In order to compare the DS results in a more comprehensive view, we also add the DS results of other forecasting models to Table 6. By comparison, we find that the DS results of the forecasting model using one single intelligent technology are similar to the results of our work, while all of them float around 0.5, such as LSTM in Ref. [4], BPNN, STNN, and SVM in Ref. [14]. However, the results of the hybrid forecasting model using multiple intelligent technologies have a huge advancement, while all of them are greater than 0.7, such as CEEMD-PCA-LSTM in Ref.

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Table 6. The DS results of our work and comparison results with other works.

Work	SHSECI	SZSECI	NASDAQCI	LFTSE100I
Our Work	0.5282	0.5242	0.5078	0.5097
RNN in Ref. [4]	0.4769	0.4867	N/A	N/A
LSTM in Ref. [4]	0.4631	0.5015	N/A	N/A
CEEMD-PCA-LSTM in Ref. [4]	0.7345	0.7443	N/A	N/A
BPNN in Ref. [14]	N/A	N/A	N/A	0.5299
STNN in Ref. [14]	N/A	N/A	N/A	0.5331
SVM in Ref. [14]	N/A	N/A	N/A	0.5283
EMD-STNN in Ref. [14]	N/A	N/A	N/A	0.7507

[4] and EMD-STNN in Ref. [14]. We speculate on the reason behind this phenomenon that the forecasting model based on single intelligent technology lacks the ability of feature extraction and analysis, which is not as good as the hybrid forecasting model based on multiple intelligent technologies. Hence, our forecasting system has certain direction trend prediction ability, yet it is weaker than the hybrid prediction system.

5. Transfer learning and generalization ability

5.1. Transfer learning in stock sector

Stocks with short issuance time and few available data are often encountered in stock market forecasting. It is difficult for the forecasting system to predict these stocks with little data effectively by utilizing the previous training and testing methods. Nevertheless, transfer learning, which applies the learned knowledge in the source domain to the new target domain, can skillfully solve the above difficult problem, since the stock price changes of companies in the same stock sector and the stock sector itself often have correlation and linkage.

In this work, the capacity of transfer learning capacity in nanophotonic reservoir computing system based on silicon OMO is explored by exploiting the massive data accumulated for a long time in the agricultural service stock sector (ASSS, SS: 881104) to forecast the stock prices of four companies with short listing time under this sector. The four companies under ASSS are Tianjin Ringpu Bio-technology Co.,Ltd. (RPBT, SS: 300119), Dayu Water-Saving Group Co.,Ltd. (DYWS, SS: 300021), Jinhe Biotechnology Co.,Ltd. (JHBT, SS: 002688), and Wuhan Keqian Biology Co.,Ltd. (KQBL, SS: 688526), respectively. The data settings for the stock sector of agricultural service is presented in Table 7, where the corresponding training-dataset are all from ASSS, and the test-dataset are all from the four stocks under ASSS.

Table 7. Data settings for the stock sector of agricultural service.

Stock	Listing Date	Training-dataset Period	Training Days	Test-dataset Period	Test Days
RPBT	17/09/2010	17/07/2009-03/08/2018	2200	06/08/2018-06/08/2021	731
DYWS	30/10/2009	17/07/2009-03/08/2018	2200	06/08/2018-06/08/2021	731
JHBT	13/07/2012	17/07/2009-03/08/2018	2200	06/08/2018-06/08/2021	731
KQBL	22/09/2020	12/02/2018-22/09/2020	635	23/09/2020-06/08/2021	211

The listing date of ASSS is 1st August 2007, which is earlier than all the four stocks shown in Table 7. Furthermore, the start dates of the training-dataset periods in Table 7 are all earlier than the respective listing date. Undoubtedly, the amount of data accumulated from the issuance of the four stocks cannot provide sufficient training data to support their respective training processes, especially the KQBL whose listing time is less than one year old. Accordingly, the transfer

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learning has vital strategic implication and practical worth to forecast stocks with short issuance time and less data, such as RPBT, DYWS, JHBT, and KQBL benefiting from ASSS. In Table 7, the test-dataset period of RPBT, DYWS, and JHBT are all set to be three years from 6th August 2018 to 6th August 2021, while the test-dataset period of KQBL is set from 23rd September 2020 to 6th August 2021. For good forecast performance, we tried different training days and finally determined it as 2200 days and 635 days in Table 7. It merits our special attention that the training set data from ASSS and the test set data from four stocks under ASSS need to be normalized separately due to the gaps between them are excessive large. In addition, the individual training and testing processes of RPBT, DYWS, JHBT, KQBL, and ASSS in this section are all set in the situation of $k_{fb} = 1\times 10^{4}$ and $k_{inj} = 1\times 10^{-15}$.

The forecasting results of four stocks under ASSS via employing transfer learning are shown in Fig. 7, where the red and blue lines represent the real values and the forecast values respectively. In Figs. 7(a)-(d), the forecast blue curves are very close to real red curves, and the forecast errors between them are quantified in Table 8. As Table 8 illustrated, all the NMSEs, RMSEs, MAEs, and MAPEs are less than 0.09, 0.06, 0.05, and 4.32×10^{-5}, respectively. In order to more intuitively reflect the practicability of the transfer learning prediction results shown in Fig. 7, we explore the percentage change of the predicted stock price compared to the previous day with rising or falling. Figure 8 demonstrate the forecast percentage error of four stocks compared to the previous day, where the red and blue lines represent the real errors and the forecast errors, respectively. It can be seen from Figs. 8(a)-(d), the forecast percentage errors with blue curves are very close to the real percentage errors with red curves, and the real and forecast percentage errors are mainly concentrated in the range of -10% to 10%. Moreover, for more accurately reflecting the difference between the daily real percentage error and the predicted percentage error in Fig. 8, we calculate the average values of the difference between them for four stocks. The average values of the difference between real and predicted percentage errors for RPBT, DYWS, JHBT, and KQBL are separately 2.9765%, 2.1095%, 2.1633%, and 2.5369%. These average errors of no more than 3% confirm that the forecast percentage errors in Fig. 8 are very close to the real percentage errors, and further illustrate the good prediction effect of transfer learning in Fig. 7. Therefore, the results of Fig. 7, Fig. 8 and Table 8 verify that our nanophotonic reservoir computing system based on silicon OMO has the capacity of transfer learning to solve the problem of stock forecasting with short listing time and few data in the stock market.

Table 8. Evaluation metrics of forecasting results for TL.

Stock	NMSE	RMSE	MAE	MAPE
RPBT	0.0102	0.0209	0.0146	1.2313×10^{-5}
DYWS	0.0474	0.0317	0.0223	1.0952×10^{-5}
JHBT	0.0288	0.0366	0.0263	2.1050×10^{-6}
KQBL	0.0840	0.0595	0.0456	4.3113×10^{-5}

5.2. Generalization ability with stocks in same region and industry

The price of a stock in the stock market is affected not only by the company itself with the development of the industry in the region, but also by other competitive stocks in the same region and industry. Hence, the stocks in the same region and industry ordinarily have the same inherent laws and coincident correlation. In this work, the generalization ability of nanophotonic reservoir computing system is investigated by training any one of the four selected stocks in the securities industry of Shanghai to forecast the other three stocks. These four stocks are from companies of East Money Information Co.,Ltd. (EMI, SS: 300059), Haitong Securities Co.,Ltd. (HTS, SS: 600837), Shanghai Chinafortune Co.,Ltd (SHCF, SS: 600621), and Everbright Securities Co.,Ltd. (EBS, SS: 601788), respectively. The data settings for the four stocks in the securities

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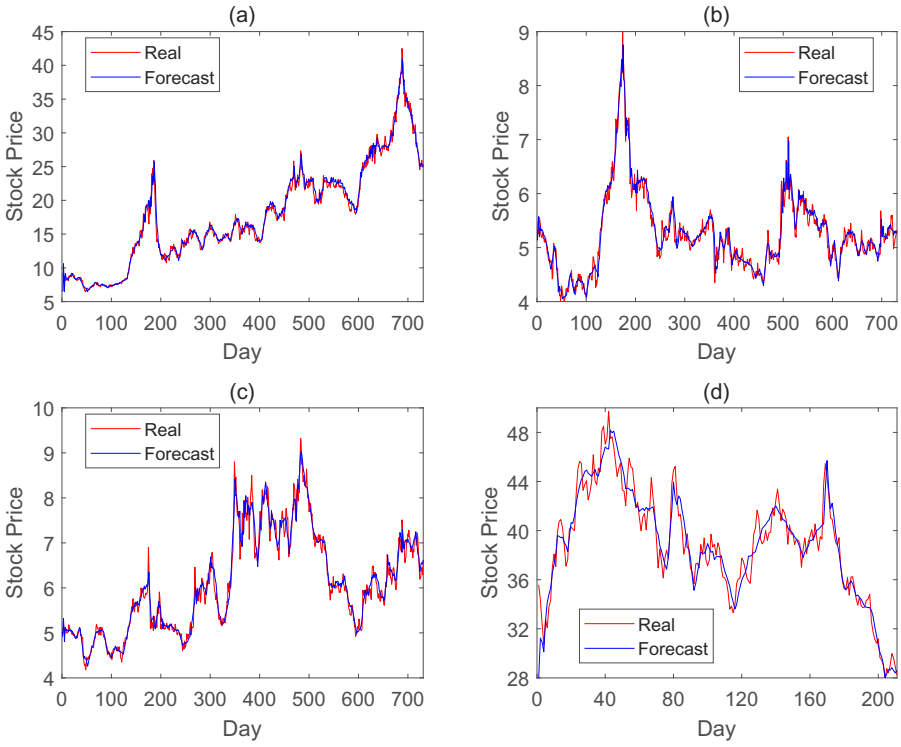


Fig. 7. The forecasting results of four stocks under ASSS by exploiting transfer learning, where the red and blue lines represent the real values and the forecast values respectively. (a) RPBT; (b) DYWS; (c) JHBT; (d) KQBL.

industry of Shanghai are presented in Table 9, where the corresponding training-dataset and the test-dataset are independent of each other in one stock. The format of "AAA-BBB" represents that the training-dataset of stock "AAA" is used for training, and the trained model is applied for testing the test-dataset of stock "BBB", where "AAA" and "BBB" denote any two different stocks in Table 9 respectively. In Table 9, the test days of four stocks are all 248 days from 3rd August 2020 to 6th August 2021. Meantime, the training days of four stocks are all 1500 days, but they have different training-dataset period, which may probably caused by the different suspension time of each stocks in the stock market. Additionally, the individual training and testing processes are all set in the situation of $k_{fb} = 1\times 10^{4}$ and $k_{inj} = 1\times 10^{-15}$.

Table 9. Data settings for four stocks in the securities industry of Shanghai.

Stock	Dataset Period	Trading Days	Training-dataset Period	Training Days	Test-dataset Period	Test Days
EMI	17/03/2014-06/08/2021	1748	17/03/2014-31/07/2020	1500	03/08/2020-06/08/2021	248
HTS	06/06/2014-06/08/2021	1748	06/06/2014-31/07/2020	1500	03/08/2020-06/08/2021	248
SHCF	06/03/2014-06/08/2021	1748	06/03/2014-31/07/2020	1500	03/08/2020-06/08/2021	248
EBS	30/05/2014-06/08/2021	1748	30/05/2014-31/07/2020	1500	03/08/2020-06/08/2021	248

The forecasting results of four stocks in the securities industry of Shanghai are presented in Fig. 9, where the red and blue lines represent the real values and the forecast values respectively. In Figs. 9(a1)-(a3), the training-dataset are all from EMI, while tested on the test-dataset of HTS, SHCF, and EBS respectively. The similar process for HTS, SHCF, and EBS as the training-dataset

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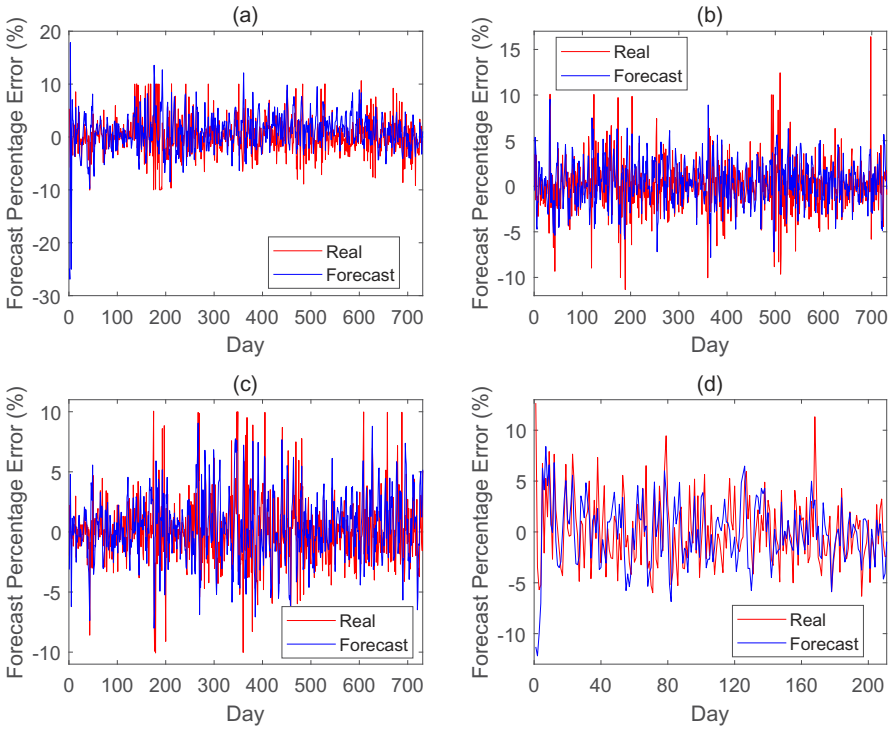


Fig. 8. The forecast percentage errors of four stocks compared to the previous day, where the red and blue lines represent the real errors and the forecast errors, respectively. (a) RPBT; (b) DYWS; (c) JHBT; (d) KQBL.

to forecast the other three stocks are also exhibited in Figs. 9(b1)-(b3), Figs. 9(c1)-(c3), and Figs. 9(d1)-(d3). The whole results of Fig. 9 show that the forecast blue curves are very close to real red curves, confirming the generalization ability of nanophotonic reservoir computing system. In addition, the quantified evaluation metrics of forecasting results corresponding to Fig. 9 is listed in Table 10. As Table 10 illustrated, all the NMSEs, RMSEs, MAEs, and MAPEs are less than 0.07, 0.07, 0.05, and 8.72×10^{-5}, respectively. Hence, the results of Fig. 9 and Table 10 corroborate that our nanophotonic reservoir computing system based on silicon OMO has the generalization ability to forecast the competitive stocks in the same region and industry in the stock market.

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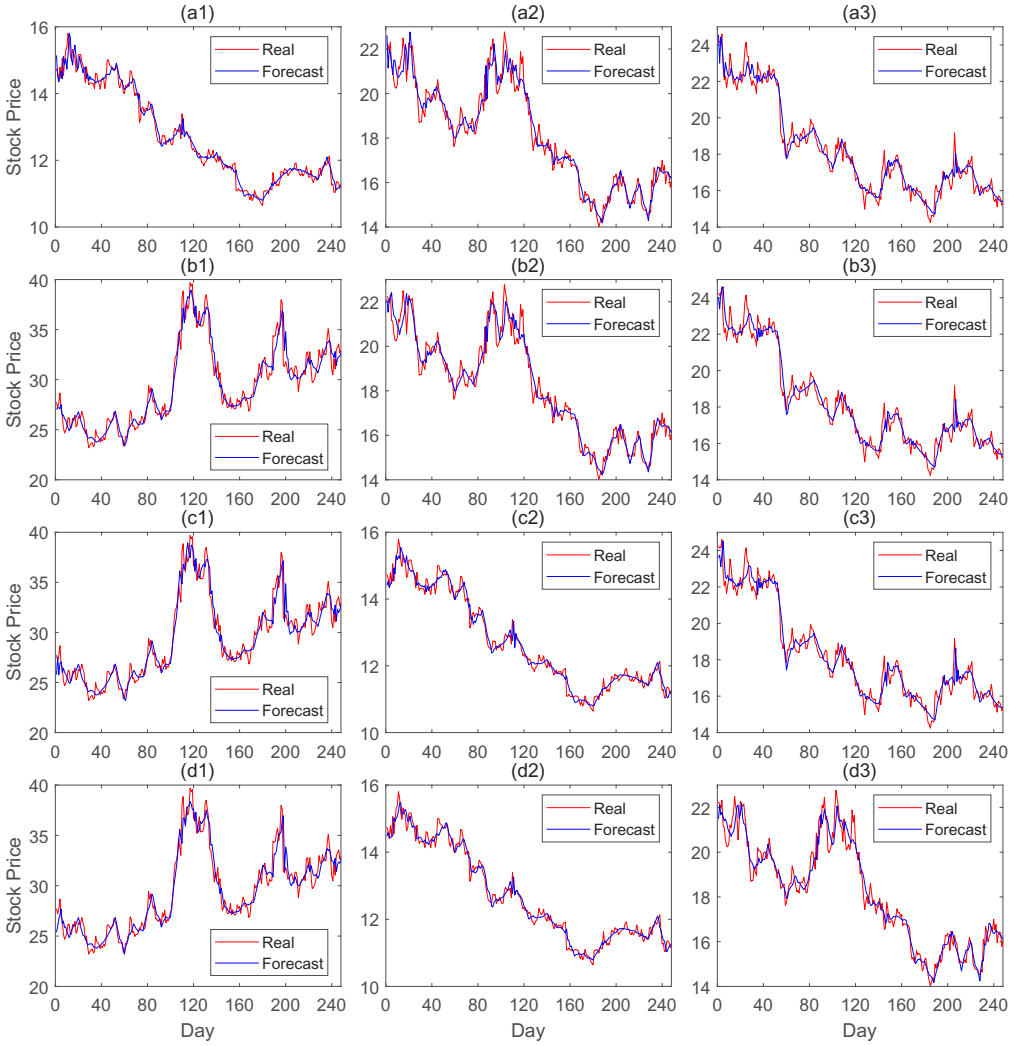


Fig. 9. The forecasting results of four stocks in the securities industry of Shanghai, where the red and blue lines represent the real values and the forecast values respectively. (a1) EMI-HTS; (a2) EMI-SHCF; (a3) EMI-EBS; (b1) HTS-EMI; (b2) HTS-SHCF; (b3) HTS-EBS; (c1) SHCF-EMI; (c2) SHCF-HTS; (c3) SHCF-EBS; (d1) EBS-EMI; (d2) EBS-HTS; (d3) EBS-SHCF.

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Table 10. Evaluation metrics of forecasting results for GA.

Stock	NMSE	RMSE	MAE	MAPE
EMI-HTS	0.0221	0.0401	0.0302	8.5573×10^{-6}
EMI-SHCF	0.0458	0.0576	0.0445	3.6631×10^{-5}
EMI-EBS	0.0396	0.0492	0.0367	2.4038×10^{-5}
HTS-EMI	0.0614	0.0603	0.0457	6.3319×10^{-5}
HTS-SHCF	0.0434	0.0561	0.0421	2.0297×10^{-5}
HTS-EBS	0.0434	0.0576	0.0380	2.2189×10^{-5}
SHCF-EMI	0.0667	0.0629	0.0473	6.6066×10^{-5}
SHCF-HTS	0.0207	0.0388	0.0299	2.7996×10^{-5}
SHCF-EBS	0.0431	0.0514	0.0383	3.1258×10^{-5}
EBS-EMI	0.0639	0.0616	0.0458	8.7171×10^{-5}
EBS-HTS	0.0196	0.0378	0.0288	3.9832×10^{-5}
EBS-SHCF	0.0417	0.0549	0.0429	4.7934×10^{-5}

6. Conclusion

In this work, we implement a novel nanophotonic reservoir computing system based on silicon OMO using PhC cavities for stock market forecasting. The nanophotonic reservoir computing system based on silicon OMO with time-delayed feedback mechanism and DWT denoising is successfully established for the first time. The long-term closing prices of SHSECI, SZSECI, NASDAQCI, and LFTSE100I are precisely forecast with tiny NMSE, RMSE, MAE, and MAPE, revealing the stock forecast of mature stock market and emerging stock market has disparate characteristics. Meanwhile, the comparison results show that our system can achieve high-quality stock index prediction as the listed literature. The influence of concept drift phenomenon on forecasting performance in stock market is investigated by changing training days, and we find that the good prediction performance prefers to appear in the range of 2000 to 4000 training days for the fixed 1000 test days. Moreover, the overall impact of RC parameters on forecasting performance are exhibited through the mapping diagrams of forecast errors in the parameter space (ξ_w, P_{in}) and (k_{inj}, k_{fb}), where the interesting variation processes are resemble to the coastline and archipelago in the geographic map. The results of these two mapping diagrams also illustrate the different variation regulations for mature stock market and emerging stock market, while the mature stock markets are more sensitive to the change of RC parameters than the emerging stock markets. The direction trend forecasting results show that our direction forecasting performance is similar to the prediction system based on single intelligence technology in the literature, and miserably worse than the hybrid prediction system. In addition, the four stocks RPBT, DYWS, JHBT, and KQBL with short issuance time and few data from ASSS are effectively forecast via transfer learning, which indicate the transfer learning capacity of our nanophotonic reservoir computing system. The other four stocks EMI, HTS, SHCF, and EBS in the securities industry of Shanghai are availably forecast with each other, revealing the generalization ability of our nanophotonic reservoir computing system.

Although the above summary illustrates the achievements of this work, there are still some limitations in this work that need to be investigated in the future. In the first instance, explore optical means that can improve the feature extraction and analysis ability of the nanophotonic reservoir computing system, or cooperate with other optical computing systems to construct hybrid forecasting systems, such as the optical neural networks (ONNs) based on programmable nanophotonic processor with deep learning application potential [49] and the integrated photonic tensor core based on optical soliton frequency comb with parallel convolutional processing

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capacity [50]. The second aspect is that the present dataset only covers the closing prices of four stock indices, one stock sector, and eight individual stocks. It is interesting to study more regionally representative stock indexes around the world with its open, high, and low prices etc. Meantime, the more individual stocks in diverse stock sectors, industries, and regions are waiting to explore more capabilities of the RC system and discover more meaningful financial phenomena. Besides, the research object can not only be limited to the stock market, but also focus on the futures market, option market, bond market, fund market, foreign exchange market and etc. The third facet is that with the improvement of feature extraction and analysis ability, more relevant information sources can be utilized to enhance the prediction ability of the stock market, such as financial news, macroeconomic factors, corporate disclosure news, investors sentiments, and even Internet social platform information. The final aspect is to improve the forecasting accuracy of stock market direction trend and construct practical trading strategies for portfolio management, which is the ultimate goal and challenge for all the investors. Desirably from these research advices to inspire scholars for splendid and wide-vision works in the future.

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